VAR:

X_1: QUANTITÁ DI COMPOSTO X ESPRESSO IN QUINTALI
X_2: " " Y " "

VINCOLI:

- $X_1 + X_2 \geq 5$
- $X_2 \leq 5$
- $2X_1 + X_2 \leq 21$

LP:

$$
\begin{aligned}
\max Z:\ & 5X_1 + X_2 \\
& X_1 + X_2 \geq 5 \\
& X_2 \leq 5 \\
& 2X_1 + X_2 \leq 21
\end{aligned}
$$

FORMA STANDARD:

$$
\begin{aligned}
\min(-Z):\ & -5X_1 - X_2 \\
& X_1 + X_2 - X_3 = 5 \\
& X_2 + X_4 = 5 \\
& 2X_1 + X_2 + X_5 = 21 \\
& X_1, X_2, X_3, X_4, X_5 \geq 0
\end{aligned}
$$

0	-5	-1	0	0	0
5	1	1	-1	0	0

NO BASE, AGGIUNGO VAR. ART.

	5	0	1	0	1	0	x^a_1
	21	2	1	0	0	1	

FASE 1:

0	0	0	0	0	0	1
5	1	1	-1	0	0	1
5	0	1	0	1	0	0
21	2	1	0	0	1	0

	-5	-1	-1	1	0	0	0
x^a_1	5	1	1	-1	0	0	1
x_4	5	0	1	0	1	0	0
x_5	21	2	1	0	0	1	0

$B = \{A_6, A_4, A_5\}$

$x = (0,0,0,5,21,5)$

$\alpha = (0,0)$

$$\theta_{max} = \min\left\{\frac{5}{1}, \frac{21}{2}\right\} = 5$$

	0	0	0	0	0	0	1
x_1	5	1	1	-1	0	0	1
x_4	5	0	1	0	1	0	0

$B = \{A_1, A_4, A_5\}$

$x = (5,0,0,5,11,0)$

$\beta = (5,0)$

x_5	11	0	-1	2	0	1	-2

FASE 2:

	25	0	4	-5	0	0	5
x_1	5	1	1	-1	0	0	1
x_4	5	0	1	0	1	0	0
x_3	11	0	-1	2	0	1	-2

$$\theta_{max} = \min\left\{\frac{11}{2}\right\} = \frac{11}{2}$$

	$\frac{105}{2}$	0	$\frac{3}{2}$	0	0	$\frac{5}{2}$	0
x_1	$\frac{21}{2}$	1	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0
x_4	5	0	1	0	1	0	0
x_3	$\frac{11}{2}$	0	$-\frac{1}{2}$	1	0	$\frac{1}{2}$	-1

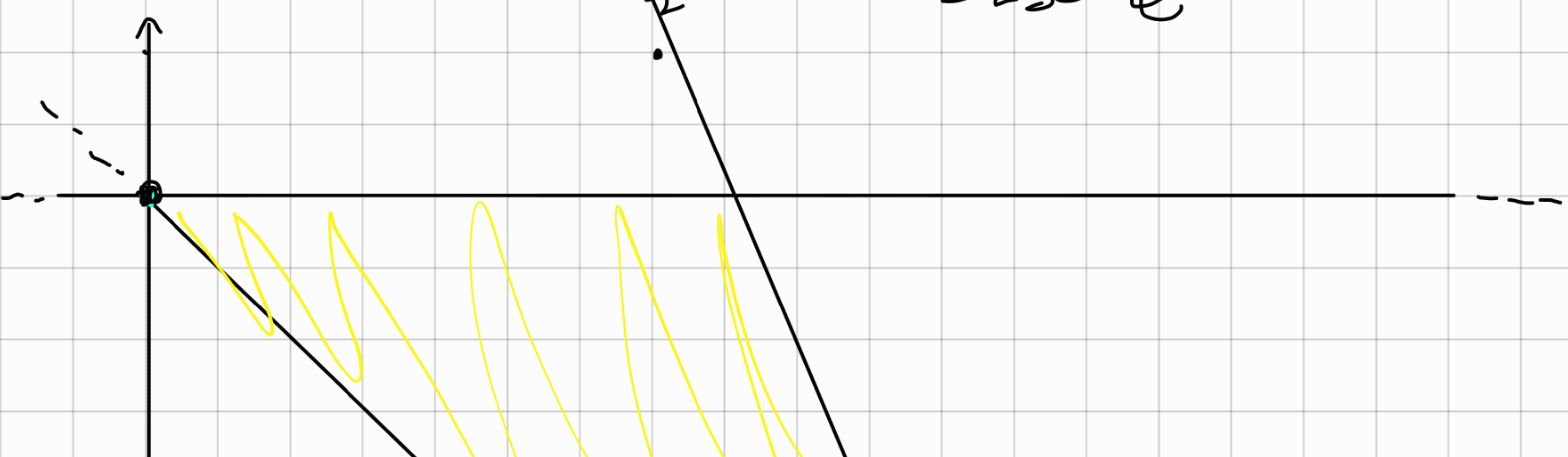
$B = \{A_1, A_4, A_3\}$

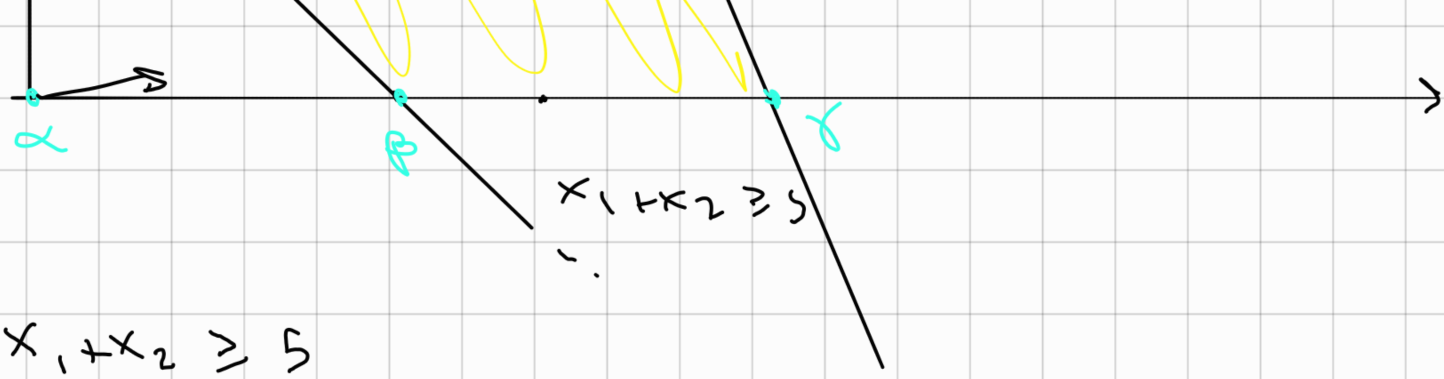
$x = \left(\frac{21}{2}, 0, \frac{11}{2}, 5, 0, 0\right)$

$f = \left(\frac{21}{2}, 0\right)$

SOL OTTIMA. PRODURRE $\frac{21}{2}$ q. di X_1 e 0 di X_2

PROFITTO $= \frac{105}{2} \cdot 100 = 5250$ €





$$x_1 + x_2 \geq 5$$
$$x_2 \leq 5$$
$$2x_1 + x_2 \leq 21$$

$$\min(-z): \; -5x_1 - x_2$$
$$\begin{aligned}
x_1 + x_2 - x_3 &= 5\\
x_2 + x_4 &= 5\\
2x_1 + x_2 + x_5 &= 21\\
x_1, x_2, x_3, x_4, x_5 &\geq 0
\end{aligned}$$

DUALE:

$$\max W: \; 5\pi_1 + 5\pi_2 + 21\pi_3$$
$$\begin{aligned}
\pi_1 + 2\pi_3 &\leq -5\\
\pi_1 + \pi_2 + \pi_3 &\leq -1\\
-\pi_1 &\leq 0\\
\pi_2 &\leq 0\\
\pi_3 &\leq 0
\end{aligned}$$

TAB. INIZ.

	-5	-1	-1	1	0	0	0
x^a_1	5	1	1	-1	0	0	1

x_4	5	0	1	0	1	0	0
x_5	21	2	1	0	0	1	0

TAB. FINALE

	$\frac{105}{2}$	0	$\frac{3}{2}$	0	0	$\frac{5}{2}$	0
x_1	$\frac{21}{2}$	1	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0
x_4	5	0	1	0	1	0	0
x_3	$\frac{11}{2}$	0	$-\frac{1}{2}$	1	0	$\frac{1}{2}$	-1

$$\bar{c}_B = c_B - \bar{c}_B \qquad c_B = (0,0,0)$$

$$\bar{c}_B = (0,0,\tfrac{5}{2})$$

$$\bar{c}_B = (0,0,-\tfrac{5}{2})$$

ILP:

$\max z: \; 5x_1 + x_2$

$x_1 + x_2 \geq 5$

$x_2 \leq 5$

$2x_1 + x_2 \leq 21$

$x_1, x_2 \in \mathbb{Z}$

TAB. FINALE

	$\frac{105}{2}$	0	$\frac{3}{2}$	0	0	$\frac{5}{2}$
x_1	21	1	1	0	0	1

x_1	[illegible]	1	[illegible]	0	0	[illegible]
x_4	5	0	1	0	1	0
x_3	$\frac{11}{2}$	0	$-\frac{1}{2}$	1	0	$\frac{1}{2}$

RIGA GEN. 1, TAGLIO GOMORY

$$\left(\frac{1}{2}-0\right)x_2 + \left(\frac{1}{2}-0\right)x_5 \geq \left(\frac{21}{2}-10\right)$$

$$\frac{1}{2}x_2 + \frac{1}{2}x_5 \geq \frac{1}{2}$$

$$x_5 = 21 - 2x_1 - x_2$$

$$\Rightarrow \cancel{\frac{1}{2}x_2} + \frac{21}{2} - x_1 - \cancel{\frac{1}{2}x_2} \geq \frac{1}{2}$$

$$-x_1 \geq \frac{1}{2} - \frac{21}{2}$$

$$x_1 \leq 10$$

TAGLIO STANDARD

$$\frac{1}{2}x_2 + \frac{1}{2}x_5 \geq \frac{1}{2} \Rightarrow -\frac{1}{2}x_2 - \frac{1}{2}x_5 + x_6 \leq -\frac{1}{2}$$

	$\frac{105}{2}$	0	$\frac{3}{2}$	0	0	$\frac{5}{2}$	0
x_1	$\frac{21}{2}$	1	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0

FATTIBILE SOLO PER DUALE

$\Rightarrow$ SIMP. DUALE

x_4	5	0	1	0	1	0	0
x_3	$\frac{11}{2}$	0	$-\frac{1}{2}$	1	0	$\frac{1}{2}$	0
x_6	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	0	$-\frac{1}{2}$	1

$$-\frac{1}{2} < 0 \Rightarrow i = 4$$

$$\theta_{MAX} = \max\left\{\frac{3}{2}(-2), \frac{5}{2}(-2)\right\} = -3$$

	51	0	0	0	0	1	3
x_1	10	1	0	0	0	0	1
x_4	4	0	0	0	1	-1	2
x_3	5	0	0	1	0	1	-1
x_2	1	0	1	0	0	1	-2

$B = \{A_1, A_4, A_3, A_2\}$ $\quad x = (10, 1, 5, 4, 0, 0)$

$\delta = (10, 1)$